Assignment 12 Solution

Q.1.A TSK fuzzy model with two rules is given as below.

$$RULE\ 1: IF\ \text{“}x\ is\ HIGH\text{”}\ AND\ \text{“}y\ is\ LOW\text{”}\ THEN\ \text{“}z\ =\ x\ +\ 2\text{”}$$

$$RULE\ 2: IF\ \text{“}x\ is\ LOW\text{”}\ AND\ \text{“}y\ is\ HIGH\text{”}\ THEN\ \text{“}z\ =\ y\ +\ 1\text{”}$$

The input membership functions are defined as follows:

$$x(LOW)\ =\ triangle(x; [1,3,5])$$

$$x(HIGH)\ =\ triangle(x; [2,4,6])$$

$$y(LOW)\ =\ triangle(y; [0,2,4])$$

$$y(HIGH)\ =\ triangle(y; [6,8,10])$$

For crisp inputs$x = 1$ and$y = 1$, which of the following set of rules will be applicable?

(a) Rule 2
(b) Rule 1
(c) Rule 1 and Rule 2
(d) None of the above

Solution: (d)

The triangular membership function is defined using,

$$traingle(x; [a, b, c]) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

The intersection of crisp inputs with input membership functions are:

$$x(LOW)|_{x=1} = triangle(x; [1,3,5]) = \max\left(\min\left(\frac{1-1}{3-1}, \frac{5-1}{5-3}\right), 0\right) = \max(\min(0.0,2), 0) = \max(0.0,0) = 0$$

$$x(HIGH)|_{x=1} = triangle(x; [2,4,6]) = \max\left(\min\left(\frac{1-2}{4-2}, \frac{6-1}{6-4}\right), 0\right) = \max(\min(-0.5,2.5), 0) = \max(-0.5,0) = 0$$

$$y(LOW)|_{y=1} = triangle(y; [0,2,4]) = \max\left(\min\left(\frac{1-0}{2-0}, \frac{4-1}{4-2}\right), 0\right) = \max(\min(0.5,1.5), 0) = \max(0.5,0) = 0.5$$

$$y(HIGH)|_{y=1} = triangle(y; [6,8,10]) = \max\left(\min\left(\frac{1-6}{8-6}, \frac{10-1}{10-8}\right), 0\right) = \max(\min(-2.5,4.5), 0) = 0$$

The applicable combinations of rule will be:

$IF\ \text{“}y\ is\ LOW\text{”}$ ⇒ No such rule is present; hence the option (d) is the correct answer.

Q.2. For TSK model defined in Q.1., which of the following set of rules will be applicablefor crisp inputs$x = 2$ and$y = 7$?

(a) Rule 2
(b) Rule 1

(c) Rule 1 and Rule 2
(d) None of the above

Solution: (a)

The triangular membership function is defined using,

$$traingle(x; [a, b, c]) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

The intersection of crisp inputs with input membership functions are:

$$x(LOW)|_{x=2} = triangle(x; [1,3,5]) = \max\left(\min\left(\frac{2-1}{3-1}, \frac{5-2}{5-3}\right), 0\right) = \max(\min(0.5,1.5), 0)$$
$$= \max(0.5,0) = 0.5$$

$$x(HIGH)|_{x=2} = triangle(x; [2,4,6]) = \max\left(\min\left(\frac{2-2}{4-2}, \frac{6-2}{6-4}\right), 0\right) = \max(\min(0,2), 0) = \max(0,0)$$
$$= 0$$

$$y(LOW)|_{y=7} = triangle(y; [0,2,4]) = \max\left(\min\left(\frac{7-0}{2-0}, \frac{4-7}{4-2}\right), 0\right) = \max(\min(3.5, -1.5), 0) = 0$$

$$y(HIGH)|_{y=7} = triangle(y; [6,8,10]) = \max\left(\min\left(\frac{7-6}{8-6}, \frac{10-7}{10-8}\right), 0\right) = \max(\min(0.5,1.5), 0) = 0.5$$

The applicable combinations of rule will be:

IF "x is LOW" AND "y is $HIGH$" ⇒ Rule 2 is applicable.

Q.3. For the applicable rules obtained using given crisp inputs in Q.1., what will be the firing strength using max-min composition for TSK model defined in Q.1.?

(a) 0.50
(b) 0.20
(c) 0.25
(d) None of the above

Solution: (d)

No rule is applicable.

Q.4. For the applicable rules obtained using given crisp inputs in Q.2., what will be the firing strength using max-min composition for TSK model defined in Q.1.?

(a) 0.50
(b) 0.20
(c) 0.25
(d) None of the above

Solution: (a)

Since the applicable rule is: $RULE\ 2$: IF "x is LOW" AND "y is $HIGH$" $THEN$ "$z = y + 1$"

$$w_1 = x(LOW)|_{x=2} = 0.5$$

$$w_2 = y(HIGH)|_{y=7} = 0.5$$

The firing strength will be obtained using max-min composition as,

$$w = \min(w_1, w_2) = \min(0.5,0.5) = 0.50$$

Q.5. For the applicable rule obtained using given crisp inputs in Q.2., what will be the firing strength using max-product composition for TSK model defined in Q.1.?

(a) 0.50
(b) 0.20
(c) 0.25
(d) None of the above

Solution: (c)

Since the applicable rule is: $RULE\ 2: IF\ "x\ is\ LOW"\ AND\ "y\ is\ HIGH"\ THEN\ "z\ =\ y\ +\ 1"$

$$w_1 = x(LOW)|_{x=2} = 0.5$$

$$w_2 = y(HIGH)|_{y=7} = 0.5$$

The firing strength will be obtained using max-product composition as,

$$w = w_1 \times w_2 = 0.5 \times 0.5 = 0.25$$

Q.6. For TSK model defined in Q.1., which of the following set of rules will be applicable for crisp inputs $x = 5$ and $y = 3$?

(a) Rule 2
(b) Rule 1
(c) Rule 1 and Rule 2
(d) None of the above

Solution: (b)

The triangular membership function is defined using,

$$traingle(x; [a, b, c]) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

The intersection of crisp inputs with input membership functions are:

$$x(LOW)|_{x=5} = triangle(x; [1,3,5]) = \max\left(\min\left(\frac{5-1}{3-1}, \frac{5-5}{5-3}\right), 0\right) = \max(\min(2,0), 0) = \max(0.5,0)$$
$$= 0$$

$$x(HIGH)|_{x=5} = triangle(x; [2,4,6]) = \max\left(\min\left(\frac{5-2}{4-2}, \frac{6-5}{6-4}\right), 0\right) = \max(\min(1.5,0.5), 0)$$
$$= \max(0.5,0) = 0.5$$

$$y(LOW)|_{y=3} = triangle(y; [0,2,4]) = \max\left(\min\left(\frac{3-0}{2-0}, \frac{4-3}{4-2}\right), 0\right) = \max(\min(1.5,0.5), 0) = 0.5$$

$$y(HIGH)|_{y=3} = triangle(y;[6,8,10]) = \max\left(\min\left(\frac{3-6}{8-6},\frac{10-3}{10-8}\right),0\right) = \max(\min(-1.5,3.5),0) = 0$$

The applicable combinations of rule will be:

IF "x is $HIGH$" AND "y is LOW" ⇒Rule 1 is applicable.

Q.7. For the applicable rule obtained using given crisp inputs in Q.6., what will be the firing strength using max-min composition?

(a) 0.75
(b) 0.00
(c) 0.50
(d) 0.25

Solution: (c)

Since the applicable rule is: $RULE\ 1$: IF "x is $HIGH$" AND "y is LOW" $THEN$ "$z\ =\ x\ +\ 2$"

The firing strength will be obtained using max-min composition as,

$$w = \min(w_1, w_2) = \min(0.5,0.5) = 0.50$$

Q.8. For the applicable rule obtained using given crisp inputs in Q.6., what will be the firing strength using max-product composition?

(a) 0.75
(b) 0.00
(c) 0.50
(d) 0.25

Solution: (d)

Since the applicable rule is: $RULE\ 1$: IF "x is $HIGH$" AND "y is LOW" $THEN$ "$z\ =\ x\ +\ 2$"

The firing strength will be obtained using max-product composition as,

$$w = w_1 \times w_2 = 0.5 \times 0.5 = 0.25$$

Q.9. For the applicable rule obtained using given crisp inputs in Q.6., what will be aggregated weighted output using max-min composition?

(a) 7
(b) 3
(c) 2
(d) 4

Solution: (a)

The aggregated weighted output is calculated using,

$$z^* = \frac{w \times z}{w} = z$$

Since the applicable rule for given crisp inputs $x = 5$ and$y = 3$ is:

$$RULE\ 1: IF\ \text{“}x\ is\ HIGH\text{”}\ AND\ \text{“}y\ is\ LOW\text{”}\ THEN\ \text{“}z\ =\ x\ +\ 2\text{”}$$

$$z^* = z = x + 2 = 5 + 2 = 7$$

Q.10. For the applicable rule obtained using given crisp inputs in Q.2., what will be aggregated weighted output using max-min composition?

(a) 7
(b) 3
(c) 12
(d) 8

Solution: (d)

The aggregated weighted output is calculated using,

$$z^* = \frac{w \times z}{w} = z$$

Since the applicable rule for given crisp inputs $x = 2$ and$y = 7$ is:

$$RULE\ 2: IF\ \text{“}x\ is\ LOW\text{”}\ AND\ \text{“}y\ is\ HIGH\text{”}\ THEN\ \text{“}z\ =\ y\ +\ 1\text{”}$$

$$z^* = z = y + 1 = 7 + 1 = 8$$